

COP2805C/COP3330C Module 11 Programming Assignment

In this assignment, we will work with a Java program that demonstrates localization and resource management using exceptions. This program will implement the Command pattern, a design pattern that encapsulates each request as an object, allowing different actions—such as displaying messages and loading properties—to be executed independently with a consistent interface.

The starter code provides a program which loads and displays localized messages using resource bundles for English and French. The Command pattern implementation uses command classes (DisplayMessagesCommand and LoadPropertiesCommand) which define an execute method, allowing the main program to invoke these actions, which enhances the application's modularity and flexibility.

Instructions

Add support for German localization by adding a MessagesBundle_de_DE.properties file with German translations for greeting, farewell, application name, and error messages to the ResourceBundle provided (content for this file provided below).

Then refactor the LoadPropertiesCommand class's **execute()** method to replace its standard try/catch block with a try-with-resources statement for handling file input.

Submission

Submit your solution with the German locale selected and a screenshot of your execution output to the GitHub Classroom repository for this assignment.

Here is the required content for the German MessagesBundle_de_DE.properties file ("de" stands for **Deutsch**, the German language code according to ISO 639-1 standards).

```
greeting=Hallo!
farewell=Auf Wiedersehen!
appName=Anwendungsname
error=Fehler
```

Extra credit opportunity! I will award up to 10 points of extra credit for adding one additional locale/properties file to your solution. Points will be awarded as specified for only **one** of the choices shown below. You will need to include a screenshot of your execution output using your chosen locale (hint: the IntelliJ IDE output will support alternate display characters; other IDEs may not; the Windows command line will not unless you have alternative language packs loaded).

10-Point (Complex) Locale Choices

1. **Mandarin Chinese (Simplified)** - MessagesBundle_zh_CN.properties
 - Language Code: zh (Chinese), Country Code: CN (China)
 - This locale requires Unicode support for characters and is widely spoken.
2. **Arabic** - MessagesBundle_ar_SA.properties
 - Language Code: ar (Arabic), Country Code: SA (Saudi Arabia)
 - Arabic presents complexity with right-to-left (RTL) text direction and unique character shapes.

5-Point (Simpler) Locale Choices

1. **Spanish (Spain)** - MessagesBundle_es_ES.properties
 - Language Code: es (Spanish), Country Code: ES (Spain)
 - Spanish is a widely used Latin-based language and typically easier to integrate than non-Latin scripts.
2. **Italian (Italy)** - MessagesBundle_it_IT.properties
 - Language Code: it (Italian), Country Code: IT (Italy)
 - Italian is also Latin-based and closely resembles English in structure, making it simpler to add.

Here are some resources you can use to research the required content for the extra credit part of the exercise:

1. Online Translation Tools

- **Google Translate:** Provides quick translations and is a good starting point for individual words or short phrases. Use with caution for nuanced or context-specific translations.
 - [Google Translate](#)
- **DeepL Translator:** Known for more accurate translations, especially for European languages like Spanish and Italian.
 - DeepL Translator

2. Language-Specific Resources

- **For Mandarin Chinese (Simplified):**
 - **Chinese Grammar Wiki** by AllSet Learning: Useful for understanding Mandarin sentence structure and common expressions.
 - Chinese Grammar Wiki
 - **Pleco Dictionary App:** A reputable Chinese-English dictionary, helpful for verifying the meanings of translated words.
 - [Pleco](#)
- **For Arabic:**
 - **Arabic Language Learning on Omniglot:** Provides Arabic greetings, farewells, and other common expressions with explanations.
 - Omniglot Arabic
 - **Almaany Arabic Dictionary:** An Arabic-English dictionary useful for precise terminology.
 - Almaany Dictionary

3. Localization Standards and Phrase Guides

- **Microsoft Language Portal:** A database of localized terminology from Microsoft, including phrases in Spanish, Italian, Arabic, and more.
 - [Microsoft Language Portal](#)
- **i18n Phrases and Resources:**
 - **Phrases for Internationalization (i18n):** Lists common phrases used in apps, like "Welcome" and "Error," with translations in multiple languages.
 - Phrase.com i18n Guide

- **W3C Internationalization (i18n) Resources:** Guidelines and considerations for creating multilingual content, covering languages with unique characteristics like Arabic and Mandarin.
 - [W3C i18n](#)